

Bags, Vectors & Transformers

Bag-of-Words Approaches · Day 1, Part 4

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Three (Common) Ways to Analyze a Bag of Words

Once your text is a **document-term matrix**, what can you actually do with it? Three classic families of approaches, differing mainly in **how much they rely on the researcher versus the data**.

Dictionary

You bring a **list of words**.
The method counts them.
No learning involved.

Supervised ML

You bring **labeled examples**. The model learns to predict labels for new text.

Topic modeling

You bring **no labels**. The model finds themes on its own.

A Key Distinction: Supervised vs. Unsupervised

Supervised

- You provide **labeled data** (examples with known answers)
- The model learns the **mapping** from text to label
- Goal: **predict** labels for new, unseen text
- Example: coding tweets as pro/anti a policy

Unsupervised

- You provide **no labels**, just the text
- The model finds **structure or patterns** on its own
- Goal: **discover** themes, groupings, dimensions
- Example: finding topics in open survey responses

Dictionaries sit slightly apart: rule-based, no learning at all — but like supervised methods, you decide in advance what you are looking for.

Approach 1: Dictionary Methods

A **dictionary** (or **lexicon**) is a predefined **list of words** associated with a category. You count how often those words appear — that count becomes your measure.

The mechanics:

- ✓ Define categories and their word lists
- ✓ Count matches in each document
- ✓ Often normalize by document length
- ✓ Compare scores across documents

Example: sentiment

Positive: good, great, love, happy

Negative: bad, hate, awful, sad

"I love this, it's great"

→ 2 positive, 0 negative

Dictionary Methods: Pros & Cons

Strengths

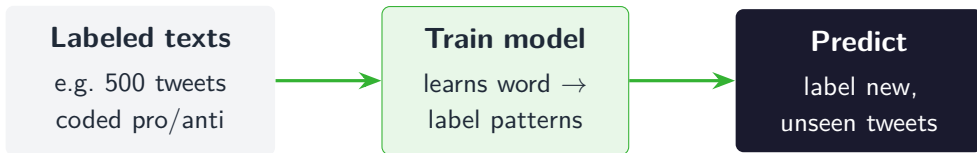
- **Transparent** — you see exactly what is counted
- **No training data** needed
- **Fast** and reproducible
- Easy to explain to reviewers
- Good when categories are well-defined

Weaknesses

- **Context-blind** — “not good” counts as positive
- Misses **sarcasm**, irony, negation
- Dictionaries are **domain-specific**
- Off-the-shelf lists may not fit your data
- Validity depends entirely on the word list

Approach 2: Supervised Machine Learning

Instead of specifying words yourself, you give the model **labeled examples** and let it **learn** which features (words) predict which label.



Common models: **Naive Bayes**, **logistic regression**, **support vector machines**. The DTM is the input; the labels are what you are predicting. You hold out some data to **validate** accuracy.

Supervised ML: Pros & Cons

Strengths

- **Learns from your data** — fits your domain
- Captures **subtle patterns** a dictionary misses
- **Measurable accuracy** via held-out data
- Scales to millions of documents once trained

Weaknesses

- Needs **labeled training data** (costly to make)
- Labels must be **reliable** and consistent
- Can **overfit** or learn spurious cues
- Less transparent — more of a “black box”

Approach 3: Topic Modeling

Topic modeling is **unsupervised**: you give it only the text, and it discovers clusters of words that tend to **co-occur** — which we interpret as **topics**. The most common method is **LDA** (Latent Dirichlet Allocation).

The core intuition:

- Each **document** is a mixture of topics
- Each **topic** is a distribution over words
- The model infers both from co-occurrence
- You **interpret and label** the topics

Example output

Topic 1: climate, emissions, policy, carbon

Topic 2: farm, crop, soil, yield

You decide these are “climate policy” and “agriculture”

Topic Modeling: Pros & Cons

Strengths

- **No labeling** required
- Great for **exploration** of large corpora
- Surfaces themes you **did not anticipate**
- Handles many topics at once
- Good first pass on unfamiliar data

Weaknesses

- You must **choose the number of topics**
- Topics can be **incoherent** or hard to label
- Results can be **unstable** across runs
- **Interpretation** is subjective
- Validation is genuinely hard

Putting It Together

	Dictionary	Supervised ML	Topic modeling
Learning	None (rule-based)	Supervised	Unsupervised
You provide	A word list	Labeled examples	Just the text
Best for	Known categories	Predicting labels	Exploring themes
Main risk	Context-blind	Needs good labels	Hard to interpret

These are not rivals — researchers often combine them. A topic model to explore, a dictionary to measure, a classifier to scale.

Which Should You Use?

- **Know exactly what you're measuring** and have clear categories? Start with a dictionary.
- **Have (or can create) labeled data** and want to predict a category? Use supervised ML.
- **Not sure what's in your corpus** and want to explore? Try **topic modeling**.

The honest answer: it depends on your **research question** and what data you have. The rest of this course gives you hands-on experience with all three, so you can make that call yourself.